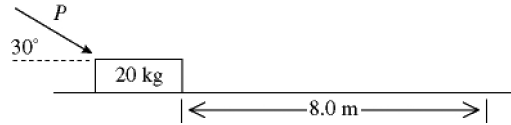


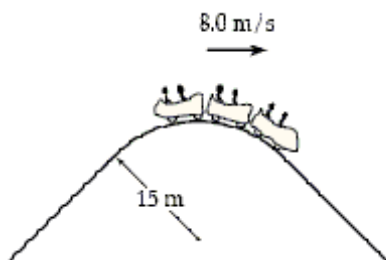
Name _____

- 1) In the figure, a constant external force $P = 160 \text{ N}$ is applied to a 20.0-kg box, which is on a rough horizontal surface. While the force pushes the box a distance of 8.00 m , the speed changes from 0.500 m/s to 2.60 m/s . The work done by friction during this process is closest to 1) _____

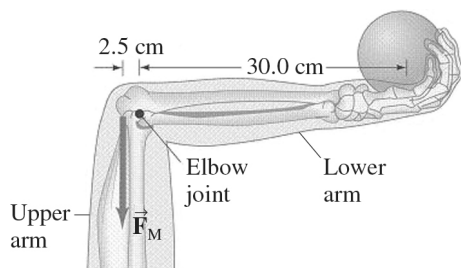


- A) -1040 J . B) $+1170 \text{ J}$. C) -1170 J . D) $+1110 \text{ J}$. E) $+1040 \text{ J}$.
- 2) A 1500-kg car accelerates from 0 to 25 m/s in 7.0 s with negligible friction and air resistance. What is the average power delivered by the engine? ($1 \text{ hp} = 746 \text{ W}$) 2) _____
- A) 90 hp B) 80 hp C) 70 hp D) 60 hp E) 50 hp
- 3) At their closest approach, Venus and Earth are $4.20 \times 10^{10} \text{ m}$ apart. The mass of Venus is $4.87 \times 10^{24} \text{ kg}$, the mass of Earth is $5.97 \times 10^{24} \text{ kg}$, and $G = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$. What is the magnitude of the gravitational force exerted by Venus on Earth at that point? 3) _____
- A) $5.43 \times 10^{26} \text{ N}$
 B) $1.10 \times 10^{18} \text{ N}$
 C) $6.30 \times 10^{20} \text{ N}$
 D) $4.62 \times 10^{28} \text{ N}$
 E) $1.72 \times 10^{19} \text{ N}$
- 4) A uniform ladder 15 ft long is leaning against a frictionless wall at an angle of 53° above the horizontal. The weight of the ladder is 30 pounds. A 75-lb boy climbs 6.0-ft up the ladder. What is the magnitude of the friction force exerted on the ladder by the floor . hint: you don't need to convert pounds to newtons. 4) _____
- A) 34 lb B) 38 lb E) 47 lb
 C) 24 lb D) 43 lb
- 5) A 480-kg car moving at 14.4 m/s hits from behind a 570-kg car moving at 13.3 m/s in the same direction. If the new speed of the heavier car is 14.0 m/s , what is the speed of the lighter car after the collision, assuming that any unbalanced forces on the system are negligibly small? 5) _____
- A) 10.5 m/s B) 19.9 m/s C) 13.6 m/s D) 5.24 m/s
- 6) An astronaut goes out for a "space-walk" at a distance above the earth equal to the radius of the earth. What is her acceleration due to gravity at that point? 6) _____
- A) $g/\sqrt{2}$ B) $g/4$ C) zero D) g E) $g/2$

- 7) A roller-coaster car has a mass of 500 kg when fully loaded with passengers. The car passes over a hill of radius 15 m, as shown. At the top of the hill, the car has a speed of 8.0 m/s. What is the force of the track on the car at the top of the hill? 7) _____



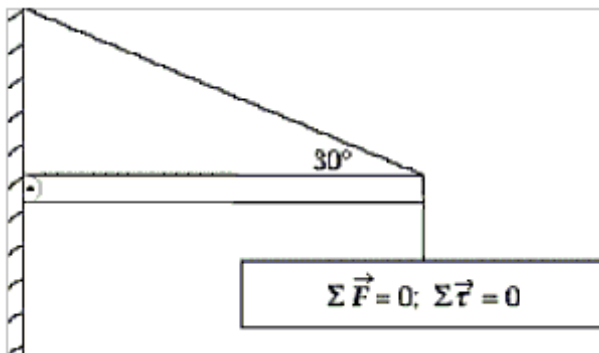
- A) 2.8 kN up
B) 7.0 kN down
C) 5.6 kN down
D) 7.0 kN up
E) 5.6 kN down
- 8) A 2.4-kg ball falling vertically hits the floor with a speed of 2.5 m/s and rebounds with a speed of 1.5 m/s. What is the magnitude of the impulse exerted on the ball by the floor? 8) _____
- A) 1.6 N.s
B) 9.6 N.s
C) 6.4 N.s
D) 2.4 N.s
- 9) An athlete holds a 7.5-kg shot put in his hand with his lower arm horizontal, as shown in the figure. His lower arm has a mass of 2.8 kg and its center of gravity (or center of mass) is 12 cm from the elbow-joint pivot. How much force must the extensor muscle (which is $\vec{F}_M$ in the figure) in the upper arm exert on the lower arm? 9) _____



- A) 1500 N
B) 500 N
C) 100 N
D) 1000 N
- 10) The work performed as a function of time for a process is given by $W = at^3$, where $a = 2.4 \text{ J/s}^3$. What is the instantaneous power output at $t = 3.7 \text{ s}$? 10) _____
- A) 207 W
B) 138 W
C) 99 W
D) 69 W
- 11) A 0.75-kg sphere is released from rest and is moving 5.0 m/s after falling 2.0 m in a viscous medium. How much work is done by the force the viscous medium exerts on the sphere during this descent? 11) _____
- A) -6.8 J
B) -2.7 J
C) -5.3 J
D) -4.6 J
E) -6.1 J

- 12) Use the figure to answer the question. The horizontal beam supporting the sign is of uniform construction and weighs 50 N. If the sign weighs 150 N, the tension in the supporting cable is

12) _____



Length of beam does not matter. take any length you want. (should cancel out)

- A) 550N B) 350N C) 300M D) 200N
- 13) When you walk, you push on the floor to the left and the floor
- A) pushes you to the right.
B) both of these simultaneously.
C) also pushes on you to the left.
D) can only wish it could push on you.
E) none of the above
- 14) A rocket moving in outer space maintains a constant acceleration (magnitude = 20 m/s^2) while ejecting fuel at a speed of 15 km/s relative to the rocket. If the initial mass of the rocket is 3000 kg , what is the magnitude of the thrust after 800 kg of fuel have been consumed?
- A) 44kN B) 52kN
C) 36kN D) 48kN E) 56kN
- 15) A pendulum is made by letting a 2.0-kg object swing at the end of a string that has a length of 1.5 m . The maximum angle the string makes with the vertical as the pendulum swings is 30° . What is the speed of the object at the lowest point in its trajectory? (take $g=10\text{m/s/s}$)
- A) 5m/s B) 3.8m/s C) 4m/s D) 2 m/s

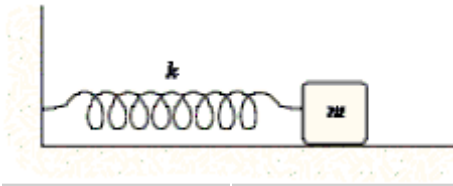
13) _____

14) _____

15) _____

- 16) The block shown is released from rest when the spring is stretched a distance d . If $k = 50$ N/m, $m = 0.50$ kg, $d = 0.10$ m, and the coefficient of kinetic friction between the block and the horizontal surface is equal to 0.25, determine the speed of the block when it first passes through the position for which the spring is unstretched.

16) _____



- A) 0.71m/s B) 0.82m/s C) 0.53m/s D) 0.61m/s

- 17) Acceleration is always in the direction:

17) _____

- A) of the final velocity B) of the displacement
C) opposite to the frictional force D) of the net force

- 18) Two ice skaters push off against one another starting from a stationary position. The 45.0-kg skater acquires a speed of 0.375 m/s. What speed does the 60.0-kg skater acquire? Assume that any other unbalanced forces during the collision are negligible.

18) _____

- A) 0.375 m/s B) 0.000 m/s C) 0.500 m/s D) 0.750 m/s E) 0.281 m/s

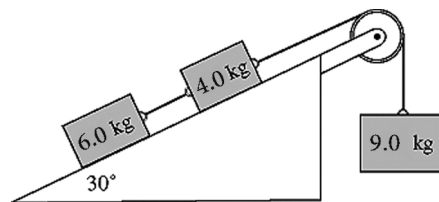
- 19) To catch a fast-moving ball, you extend your hand forward before contact with the ball and let it ride backward in the direction of the ball's motion. Doing this reduces the force of contact on your hand principally because the

19) _____

- A) force of contact is reduced.
B) time of contact is decreased.
C) time of contact is increased.
D) relative velocity is less.
E) none of the above

- 20) A system comprising blocks, a light frictionless pulley, a frictionless incline, and connecting ropes is shown in the figure. The 9.0-kg block accelerates downward when the system is released from rest. The acceleration of the system is ? about? (hint: replace the 2 blocks by just one larger one)

20) _____



- A) 0.5 m/s/s B) 3m/s/s C) 4m/s/s D) 2m/s/s E) 0m/s/s

Mechanics

$v_x = v_{x0} + a_x t$	a = acceleration
$x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$	d = distance
$v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$	E = energy
$\vec{d} = \frac{\Sigma \vec{F}}{m} = \frac{\vec{F}_{net}}{m}$	F = force
$ \vec{F}_f \leq \mu \vec{F}_n $	f = frequency
$a_c = \frac{v^2}{r}$	h = height
$\vec{p} = m\vec{v}$	I = rotational inertia
$\Delta \vec{p} = \vec{F} \Delta t$	K = kinetic energy
$K = \frac{1}{2} m v^2$	k = spring constant
$\Delta E = W = F_{\parallel} d = F d \cos \theta$	L = angular momentum
$P = \frac{\Delta E}{\Delta t}$	l = length
$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2$	m = mass
$\omega = \omega_0 + \alpha t$	P = power
$x = A \cos(\omega t) = A \cos(2\pi f t)$	p = momentum
$x_{cm} = \frac{\Sigma m_i x_i}{\Sigma m_i}$	r = radius of separation
$\alpha = \frac{\Sigma \tau}{I} = \frac{\tau_{net}}{I}$	T = period
$\tau = r_{\perp} F = r F \sin \theta$	t = time
$L = I \omega$	U = potential energy
$\Delta L = \tau \Delta t$	v = speed
$K = \frac{1}{2} I \omega^2$	W = work done on a system
$ \vec{F}_s = k \vec{x} $	x = position
	α = angular acceleration
	μ = coefficient of friction
	ϕ = angle
	τ = torque
	ω = angular speed
	$U_s = \frac{1}{2} k x^2$
	$\Delta U_g = m g \Delta y$
	$T = \frac{2\pi}{\omega} = \frac{1}{f}$
	$T_s = 2\pi \sqrt{\frac{m}{k}}$
	$T_p = 2\pi \sqrt{\frac{I}{g}}$
	$ \vec{F}_g = G \frac{m_1 m_2}{r^2}$
	$\vec{g} = \frac{\vec{F}_g}{m}$
	$U_g = -G \frac{m_1 m_2}{r}$

21)

A)

B)

C)

D)

21)

Answer Key

Testname: FINAL_23B

- 1) A
- 2) A
- 3) B
- 4) A
- 5) C
- 6) B
- 7) A
- 8) B
- 9) D
- 10) C
- 11) C
- 12) B
- 13) A
- 14) A
- 15) D
- 16) A
- 17) D
- 18) E
- 19) C
- 20) D
- 21) A